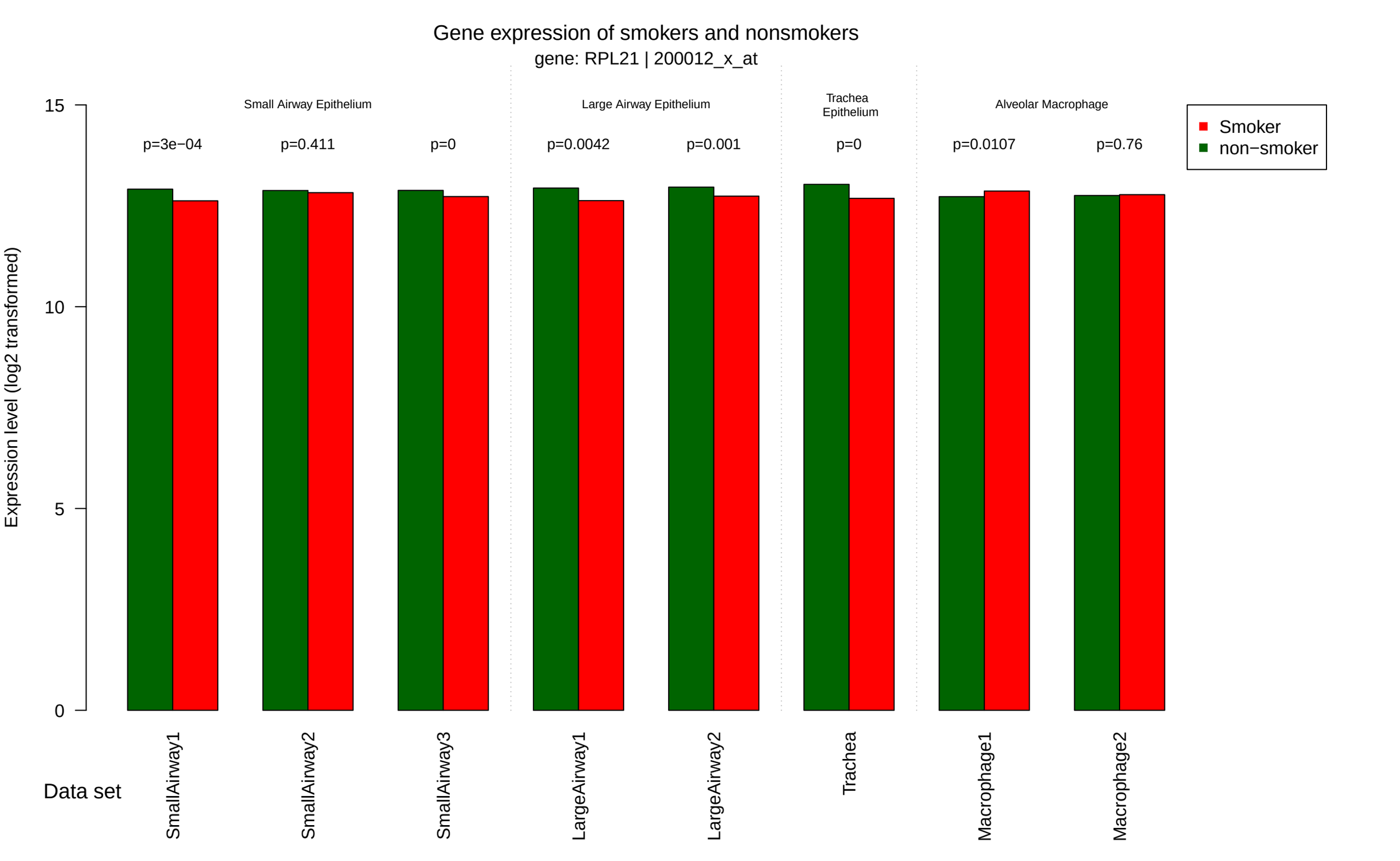
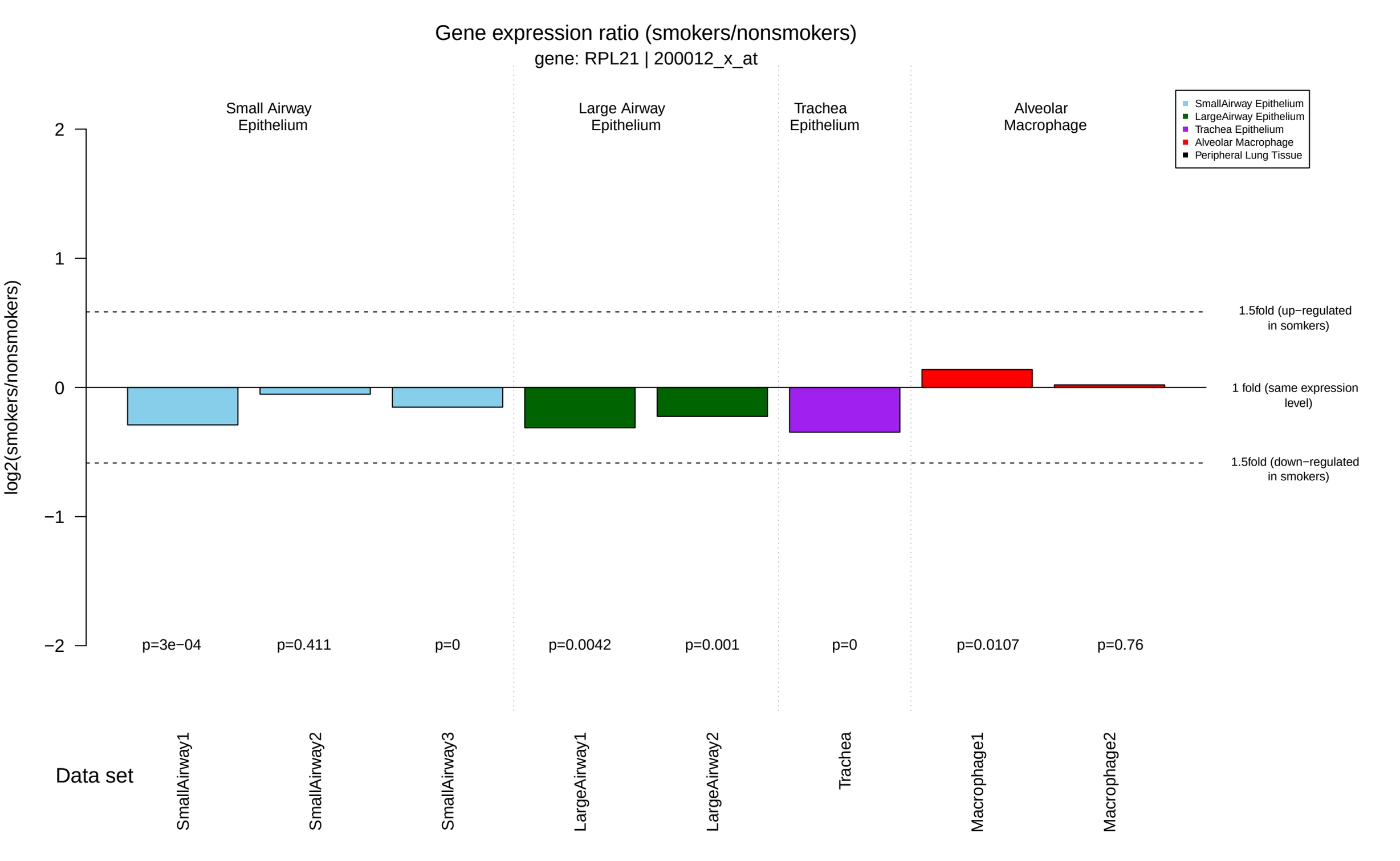
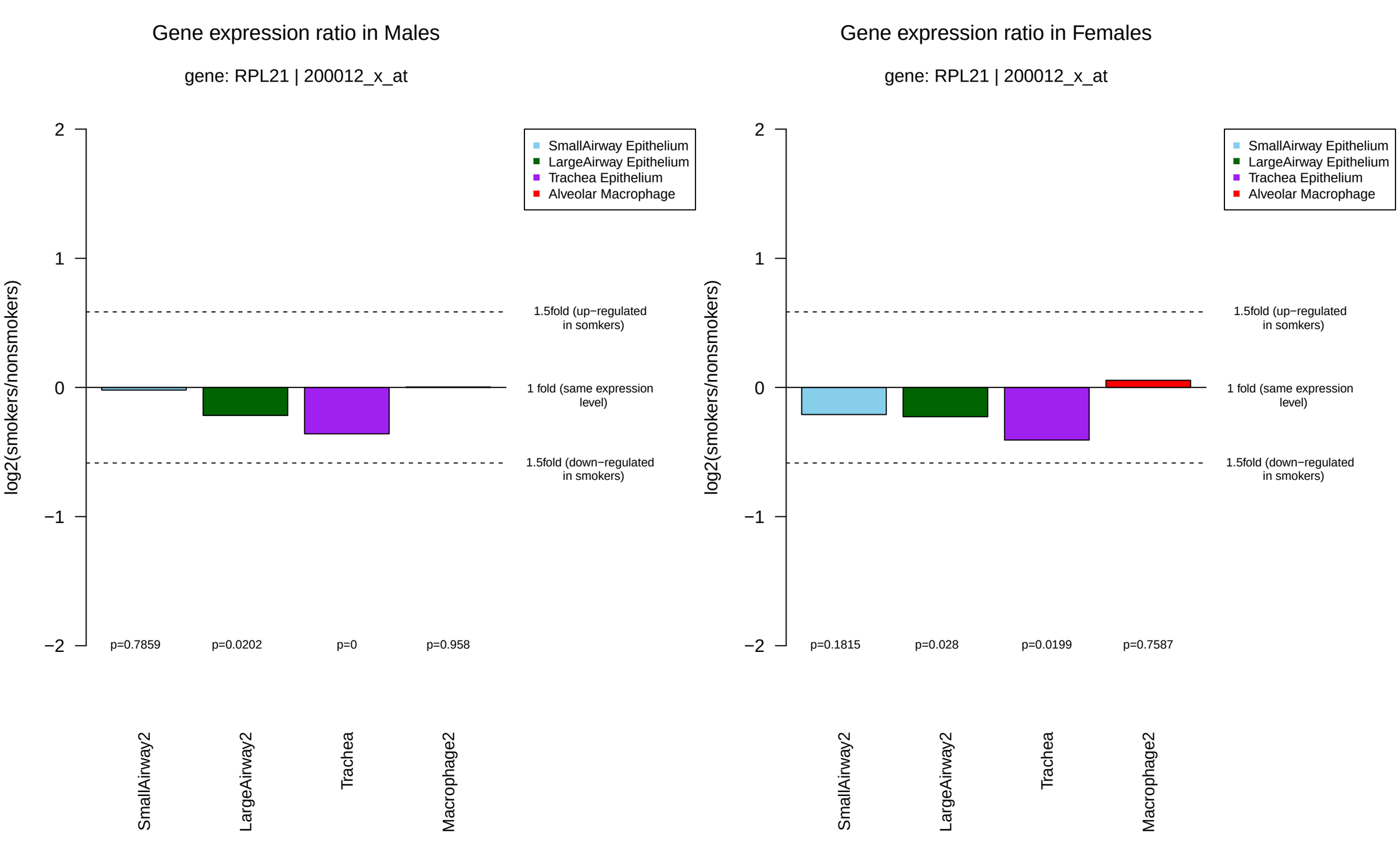
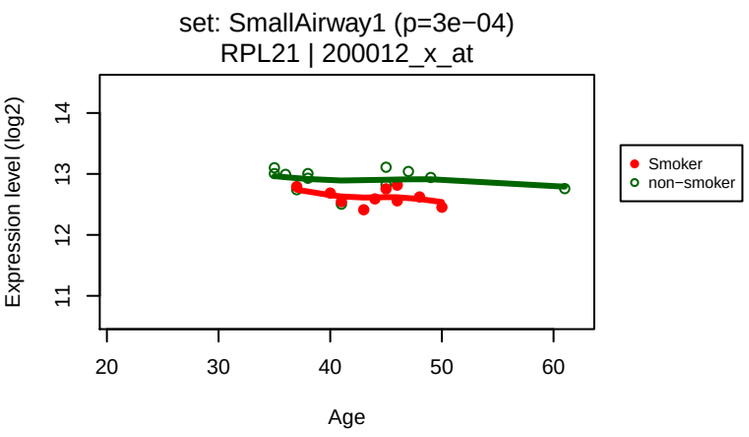
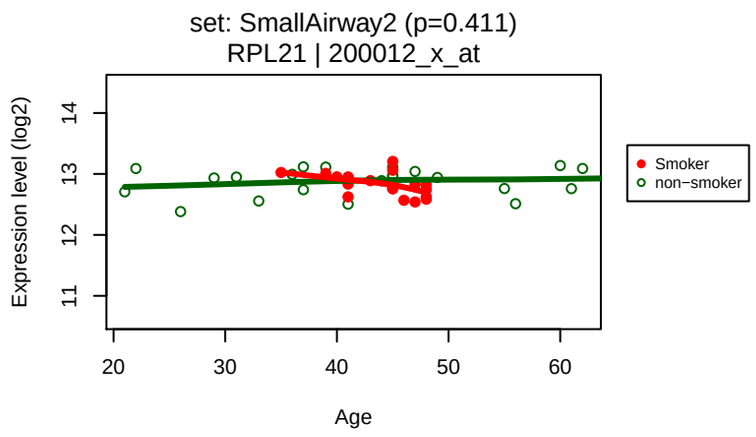
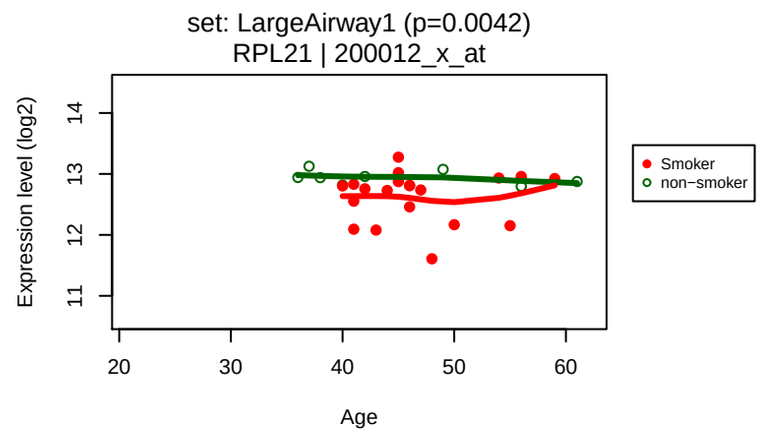

Gene expression of smokers and nonsmokers
gene: RPL21 | 200012_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=3e−04
p=0.411
p=0
p=0.0042
p=0.001
p=0
p=0.0107
p=0.76
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

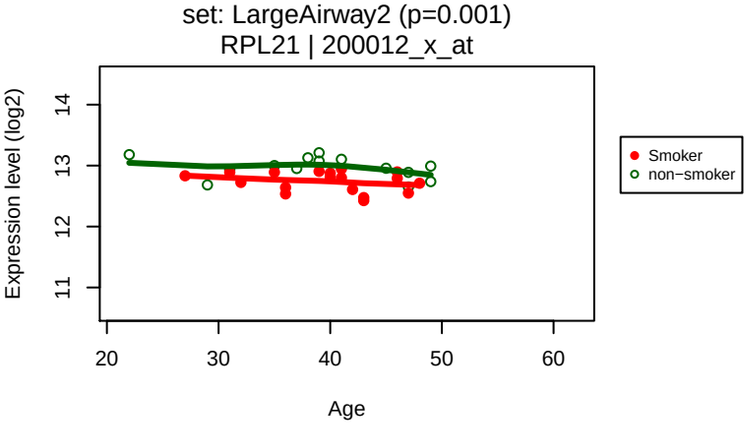

Gene expression ratio (smokers/nonsmokers)
gene: RPL21 | 200012_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=3e−04
p=0.411
p=0
p=0.0042
p=0.001
p=0
p=0.0107
p=0.76
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

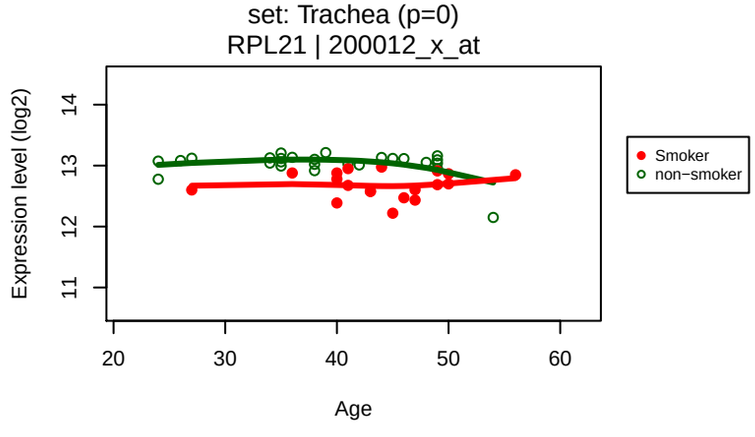

Gene expression ratio in Males
gene: RPL21 | 200012_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7859
p=0.0202
p=0
p=0.958
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RPL21 | 200012_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1815
p=0.028
p=0.0199
p=0.7587
SmallAirway2
LargeAirway2
Trachea
Macrophage2

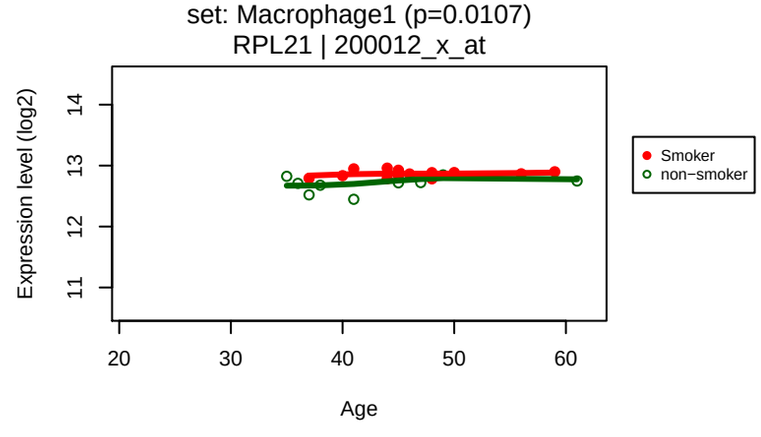

set: SmallAirway1 (p=3e−04)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker

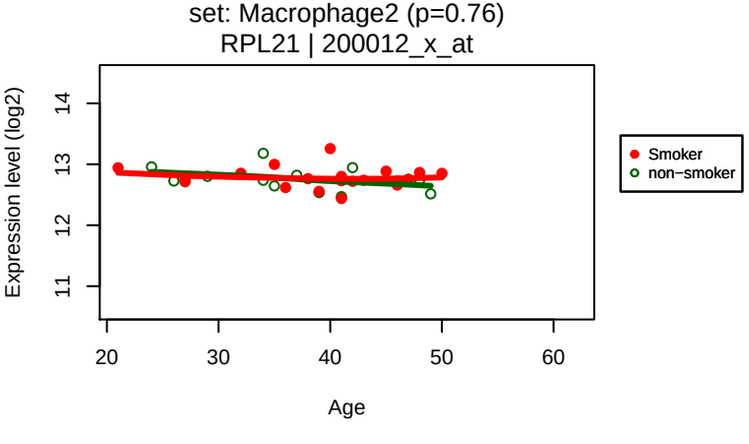

set: SmallAirway2 (p=0.411)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker

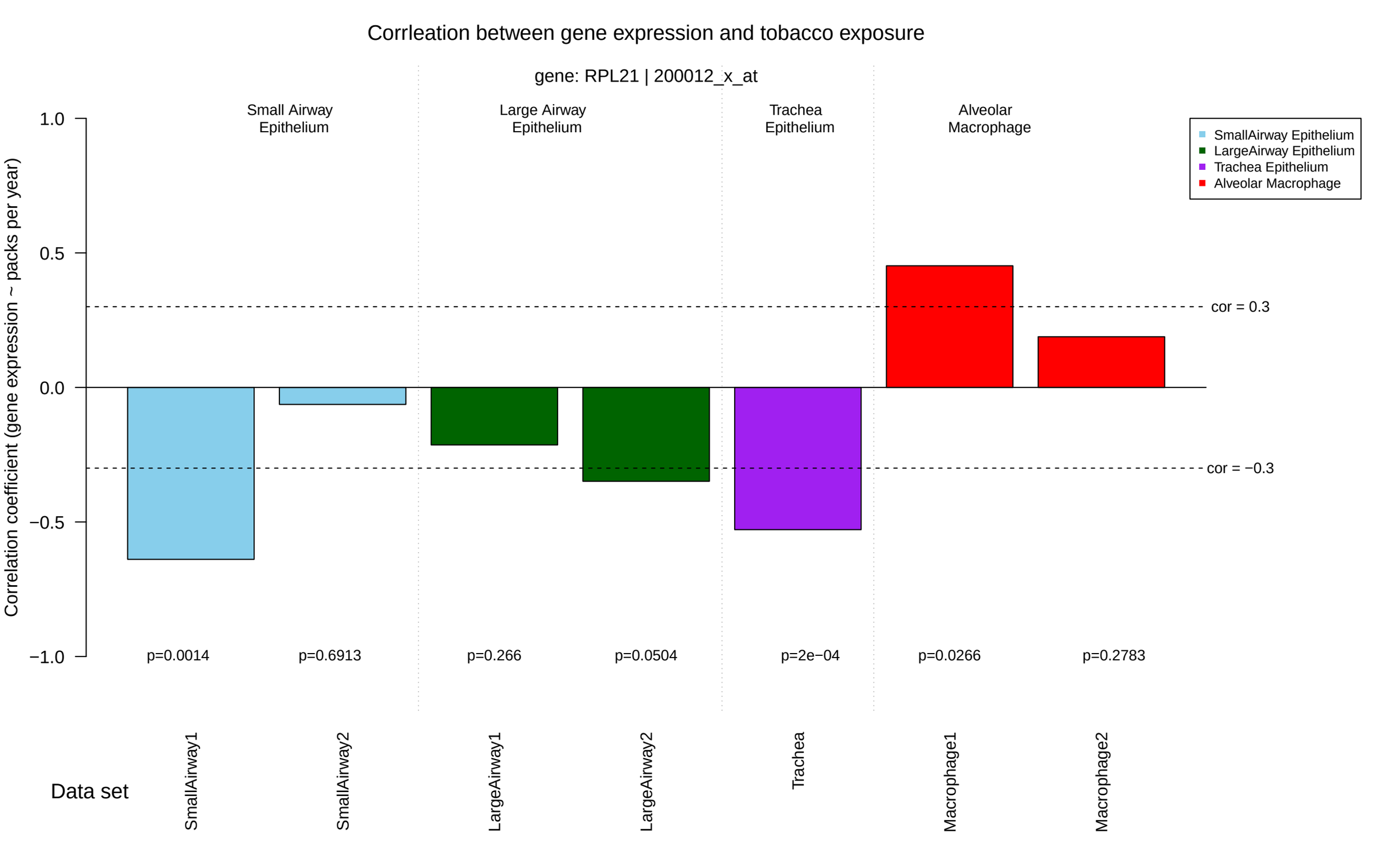

set: LargeAirway1 (p=0.0042)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.001)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0107)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.76)
RPL21 | 200012_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RPL21 | 200012_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0014
p=0.6913
p=0.266
p=0.0504
p=2e−04
p=0.0266
p=0.2783
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2